

Anindya Mondal

Researcher and Engineer

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Career Highlights

- **7 peer-reviewed publications (w/ 6 first-author):** papers in ACM TOG (SIGGRAPH Asia), AAAI, ICCV-W, and ICASSP, spanning vision, graphics, and multimodal learning.
- **4+ years of research experience:** Research Scientist/Engineer internships in enterprises like *Adobe Research*, plus doctoral research at *Surrey CVSSP* on generative and multimodal AI, with strong reproducible research engineering.
- **Teaching, mentoring & service:** Teaching Assistant for 2 MSc AI/CV modules at Surrey (2023–2025); reviewer for CVPR, ICCV, ECCV, NeurIPS, BMVC, and IEEE journals.

Area of Expertise

a) Multimodal Foundation Models (VLM, CLIP, SAM, Diffusion) - b) Controllable & Interactive Generation (Diffusion, Agentic) - c) Open-Vocabulary Object Counting - d) Large-Scale Training (DDP, FSDP, PEFT) - e) Programming (Python, C, MATLAB) - f) DL Stack (PyTorch)

Professional Experience

Research Intern, (*Adobe Research*)

Bengaluru, India 06/2026 - present

- Building rubric-aware VLM judges to evaluate generated content across semantics, spatial structure, and object relationships.
- Designing criterion-level metrics for prompt fidelity, edit correctness, layout preservation, and artifact severity.
- Refining the generation → understanding → feedback evaluation loop for generative models.

Doctoral Researcher, (*University of Surrey, CVSSP*)

Guildford, UK 10/2022 - present

- Built **ABACUS** (ACM TOG), a unified generation–understanding VLM for count-accurate, controllable image synthesis.
- Proposed **OmniCount**, cutting counting error by **56–67% MAE** on FSC-147, and released OmniCount-191 (30K images, 191 categories).
- Developed **CountLoop**, a training-free agent loop for accurate 30–200 instance generation; built reproducible PyTorch training/eval pipelines.

Research Intern, (*Indian Institute of Science*)

Bengaluru, India 05/2022 - 08/2022

- Developed a source-free domain-adaptation framework for image classification under target-domain shift.

Undergraduate Research Assistant, (*Jadavpur University*)

Kolkata, India 10/2020 - 05/2022

- Graph learning and event-based vision for signal recovery, semi-supervised segmentation, and moving-object detection.

Education

Ph.D. Artificial Intelligence *University of Surrey (CVSSP & PAI)* **Guildford, UK** 2022 - present
Research on unified vision–language models, generative AI, and object counting; first-author papers in ACM TOG, AAAI, ICCV-W; TA in 2 MSc courses; Research Intern at Adobe Research. Expected defense: late 2026.

B.E. (Hons.) Electronics & Telecom. Engineering *Jadavpur University* **Kolkata, India** 2018 - 2022
GPA 8.79/10; research on graph signal processing and event-based vision; published at ICASSP and EUSIPCO; 2 years of undergraduate research across vision and signal recovery.

Selected Publications*

- **ABACUS: Adapting a Unified Foundation Model for Bridging Image Count Understanding and Generation** *ACM Trans. of Graphics (SIGGRAPH Asia)* 2026
Anindya Mondal, Sauradip Nag, Anjan Dutta
We build a unified foundation model that bridges image count understanding and generation.
- **OmniCount: Multi-label Object Counting with Semantic–Geometric Priors** *AAAI* 2025
Anindya Mondal, Sauradip Nag, Xiatian Zhu, Anjan Dutta
We propose a training-free framework for open-vocabulary multi-label counting using semantic and geometric priors.
- **CountLoop: Training-Free High-Instance Image Generation via Iterative Agent Guidance** *Under Review* 2025
Anindya Mondal, Ayan Banerjee, Sauradip Nag, Josep Lladós, Xiatian Zhu, Anjan Dutta
We propose the first training-free high-instance image generation method using an MLLM planner–critic loop.
- **Actor-Agnostic Multi-label Action Recognition with Multi-modal Query** *ICCV-W* 2023
Anindya Mondal, Sauradip Nag, Joaquin M. Prada, Xiatian Zhu, Anjan Dutta
We propose an actor-agnostic multi-label action recognition model with multimodal queries (+50% mAP on Animal Kingdom).
- **Time-Varying Signals Recovery via Graph Neural Networks** *ICASSP* 2023
J. A. C. Correa, J. H. Giraldo, **Anindya Mondal**, et al.
We recover missing time-varying graph signals using graph neural networks.

* **Bold** = candidate; full publication list on [Google Scholar](#).

Honors and Awards

- **AAAI 2025 Grant:** Conference and travel grant for OmniCount.
- **ICCV 2023 Grant:** Conference grant.
- **Surrey Postgraduate Studentship:** Full PhD funding.
- **ACM SIGKDD India Chapter:** Uplink Research Internship Award.